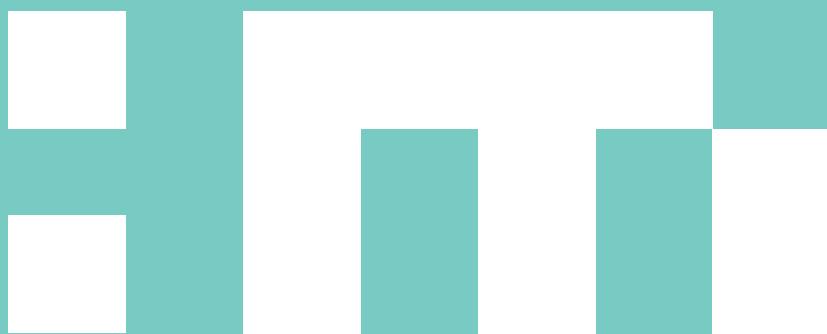


IT:U
GRADUATES
ARE
DIGITAL
TRANSFORMERS

**MASTER'S
PROGRAM**



Master of Science (MSc) ■ **Interdisciplinary Computing**

MASTER'S PROGRAM INTERDISCIPLINARY COMPUTING

Interdisciplinary Computing is the first interdisciplinary technical master's program at IT:U. Its educational approach is deeply rooted in project-based, personalized learning. As a student, you will study at the intersection of computing and various disciplines. You will work in small groups to develop your knowledge and skills in a vibrant, international environment while gaining experience for a future career in industry or academia.

At IT:U we explore the next level of learning

IT:U offers an education that combines project-based learning around real-world challenges with novel educational technologies for self-directed learning. The best way to learn new skills is to acquire them through practice within projects which address challenges that engage and motivate us.

- Take ownership of your learning by actively participating in projects.
- Collaborate in interdisciplinary groups of students and learn to approach problems from multiple perspectives.
- Participate in mission-based, real-world projects which give you hands-on experiences fostering your skills.
- Become familiar with the theories, methods, and technologies of digital transformation.
- Take courses where the main criteria for ongoing assessments are both your understanding of the subject and your ability to collaboratively solve problems.



FACTS & FIGURES

DEGREE: MSc
SEMESTERS: 4
ECTS: 120
LANGUAGE: English
LOCATION: Linz, Austria
TYPE: Full time, at your pace*
COSTS: No tuition fees**
START DATES: March and October***

Your journey through the master's program

The LEARN LAB modules will give you the opportunity to get to know the many different labs at IT:U – some of which include Virtual Reality, Motion Capture, Robotics, Data, Interaction, and more – that form the basis of IT:U's project-based, personalized learning approach. The PRE-SPECIALIZATION module will equip you with the basic skills of your chosen specialization. The RESEARCH PROFICIENCY module will train you in scientific writing and research methodology. In the SPECIALIZATION module, you will focus on real-world challenges and projects related to your specialization. The final part of your master's degree will focus on writing and finally defending your MASTER'S THESIS.

SEMESTER 1

LEARN LAB MODULE 1 30 ECTS						
Project Based Learning	Foundations and Methods 1	Digital Technologies 1	Learn Lab Project 1	Foundations and Methods 2	Digital Technologies 2	Learn Lab Project 2

SEMESTER 2

LEARN LAB MODULE 2 10 ECTS			PRE-SPECIALIZATION MODULE 20 ECTS			
Digital Technologies 3	Learn Lab Project 3	Pre-Specialization Elective A	Pre-Specialization Elective B	Pre-Specialization Project	Elective A	Elective B

SEMESTER 3

RESEARCH PROFICIENCY MODULE 10 ECTS		SPECIALIZATION MODULE 20 ECTS			
Research Proficiency		Specialization Elective A	Specialization Elective B	Interdisciplinary Seminar	Specialization Project

SEMESTER 4

MASTER'S THESIS MODULE 30 ECTS	
Master's Thesis	

* Interdisciplinary Computing at IT:U allows you to shape your own study rhythm. Courses, project work, and thesis work are organized so that students are able to balance academic work with professional and personal responsibilities.
 ** Mandatory Austrian Student Union (ÖH) fee after tolerance semesters can be found on our website.
 *** The spring intake offers a full program start, with all first-semester courses running from day one.

EXPLORE
OUR LEARN LABS!



YOUR PROFILE TO FIT WITH IT:U

- You have a bachelor's degree (180 ECTS) or equivalent in the fields of technology, natural sciences, social sciences, medicine, humanities, arts, or economics.
- You have minimum C1 level English as defined by the Common European Framework of Reference for Languages (CEFR). Please consult our FAQs page for more information on how other language assessments compare to the CEFR. Shortlisted applicants will be contacted for an interview to assess English proficiency and fit with IT:U.
- You have a strong interest in solving complex problems and working on collaborative projects.
- You are interested in combining technology and social understanding.
- You appreciate mixed methods and computational approaches, as well as exploring new AI methods.
- You enjoy a hands-on, project-based environment, where you can apply your skills to real-world challenges.
- You want to gain a broad understanding of data handling, analysis, and visualization, with a focus on practical applicability.



Example student profiles



Christina, 22 years
Bachelor in
Geoinformatics

Christina is studying geoinformatics. She loves technology and its potential for social good, especially in areas like climate change and urban sustainability. Fascinated by how data reveals patterns in society and the environment, Christina works with various data types, including social stats and sensor readings. She enjoys solving complex problems and is keen on using machine learning for environmental analysis. Christina thrives in teamwork, where she combines spatial data science with mapping to create useful insights. She aims to use her skills to support humanitarian efforts and sustainable urban planning.



Max, 22 years
Bachelor in
Computer Science

Max is skilled in programming languages like Python and Java, with a strong grasp of algorithms and data structures. He aims to specialize in solving industrial challenges by integrating IoT and AI into smart factories and automated systems. Seeking expertise in areas like process modeling and system integration, Max is ready to tackle complex problems in cross-disciplinary teams. He plans to enroll in a master's program focused on industrial software solutions and advanced manufacturing technologies to prepare for the future of industry.



Lena, 24 years
Bachelor in Physics

Lena is into using theoretical and computational models to tackle real-world challenges. She's particularly interested in the interdisciplinary blend of statistical physics, mathematics, and computer science to develop machine learning methods for understanding complex systems like climate, migration, economies, and social media. Lena aims to combine game theory, complex system modeling, network science, and AI in data-driven projects, with the goal of becoming a versatile researcher capable of solving complex problems across industry, government, and academia.



Johann, 27 years
Bachelor in Arts
& Media

Johann, a creative mind with a background in arts, is passionate about exploring new digital expressions. As a self-taught programmer and visual designer, he uses AI tools, Processing, and Arduinos in his interactive media projects. Now, with a Master in Interdisciplinary Computing, he aims to deepen his knowledge of computation to elevate his work at the intersection of design, art, and technology.

CHOOSE ONE OF FOUR SPECIALIZATIONS

1 DESIGNING INTERACTIONS

Design how people experience technology. From AI to immersive environments, learn how to create systems that are meaningful, usable, and trusted in real-world contexts. Designing Interactions focuses on how humans and technology meet. You learn how to understand human behavior, shape problems, and design interactive systems across AI, XR, robotics, and data-driven environments. Instead of only building technology, you design how it is experienced, making complex systems understandable and relevant for people.

Core LearnLabs include:

- Human-Computer Interaction Lab (INES) – understanding user behavior and interaction design
- Design Lab (DESY) – developing and prototyping design solutions
- Extended Reality Lab (RITA) – immersive AR and VR experiences

Work with emerging technologies in:

- Motion & Audio Lab (MOMO) – interaction through movement, sound, and perception
- Robotics Lab (ROB) – human-machine interaction
- Data Science Lab (DORO) – analyzing and modeling user behavior through data

Key technologies and methods include:

- AI systems, AR/VR, sensor-based systems, and human-centered data analysis

Career Prospects

Rather than preparing you for one fixed job title, this specialization equips you with a critical mindset, hands-on experience, and the ability to connect human understanding with design methods and technological expertise. Your path can take many different directions depending on your interests and projects. Graduates typically move into roles such as Product Designer, Interaction Designer, UX Researcher, Service Designer, Human Factors Engineer, Human-AI Interaction Designer, or Technology Strategist. Some also continue in academia or research as Research Assistants or PhD candidates in areas such as human-computer interaction, design, or digital technologies. These roles can be found across a wide range of industries.

“Explore possible technological futures, designing meaningful interactions with computational artifacts.”



Christopher Frauenberger
Professor of Human-Computer Interaction

2 DIGITAL EARTH, SOCIETY, AND NETWORKS

Students gain experience across all labs during the program, but for Digital Earth, Society, and Networks, the strongest links are to labs and methods that help you understand complex systems through data, space, and time.

Core LearnLabs include:

- Data Science Lab (DORO) – analyzing complex datasets, identifying patterns, and building data-driven models
- Human-Computer Interaction Lab (INES) – exploring how people interact with digital systems and data-rich environments
- Robotics Lab (ROB) – understanding intelligent systems in connected environments

Key technologies and methods include:

- Digital geospatial technologies
- Network and sensor data
- Natural language processing (NLP)
- Large language models (LLM)
- Machine learning
- Environmental measurements
- Socio-demographic statistics
- User-generated data (social media, mobile phone data, wearable sensor data, etc.)
- Computational methods for spatial and temporal analysis
- Big data analytics

Career Prospects

Digital Earth, Society, and Networks prepares you for careers where the goal is to understand complex social and environmental systems through data, technology, and spatial thinking. This specialization is especially relevant for roles such as Geospatial Data Analyst, Urban Planner, Sustainability Manager, Smart City Consultant, Mobility Planner, Geospatial Specialist, Policy Advisor, and Researcher in Digital Society and Environmental Systems.

What makes these paths attractive is that they sit at the intersection of some of today's biggest challenges: climate, cities, mobility, digital society, and public decision-making. In other words, you are not only learning how to work with data, you are learning how to use data to better understand how the world changes, how societies function, and how technology can support more informed action.

“We are focusing on understanding geospatial and social processes through the AI-based analysis of user-generated digital data.”



Bernd Resch
Professor of Geosocial Artificial Intelligence



3 DIGITAL HUMANITIES, HEALTH, AND LIFE SCIENCES

Students gain experience across all labs during the program, but for Digital Humanities, Health, and Life Sciences, the strongest links are to labs and methods that help you work with data science, data-base systems, generative AI, and data-rich systems connected to health and life sciences.

Core LearnLabs include:

- Data Science Lab (DORO) – working with complex datasets and computational methods
- Extended Reality Lab (RITA) – exploring immersive technologies that support experience, learning, and simulation

Key technologies and methods include:

- Computational methods for health, psychology, and neuroscience
- Medical data and responsible data use
- Digital tools for understanding creativity, intelligence, and human capacities
- Data-driven approaches in medical research

Digital Humanities, Health, and Life Sciences prepares you for careers where the goal is to apply computational methods to questions in health, psychology, neuroscience, and related fields. In this year's cohort, the focus is on health and life sciences, with more humanities components to be added at a later stage.



Career Prospects

This specialization is especially relevant for roles such as Digital Health Specialist, Medical Data Specialist, Health Innovation Manager, Medical Data Scientist, Neuroscience Researcher, or Research Assistant in health and life sciences. What makes these paths attractive is that they sit at the intersection of some of today's most important challenges: health, trustworthy AI, human behavior, and the responsible use of data. In other words, you are not only learning how to work with computational methods, you are learning how to apply them in ways that can improve people's lives.

“Innovation in the health and life sciences does not happen within disciplines—it happens between them.”



Spiros Denaxas
Professor of Computational Medicine

“Health data have never been richer, but turning data into actionable knowledge that improves people's lives requires both computational rigour and a deep understanding of the clinical questions that matter.”



Kristin Reiche
Professor of Digital Molecular Medicine

4 FUTURE INDUSTRIES

Students gain experience across all labs during the program, but for Future Industries, the strongest links are to labs and technologies that support industrial production, intelligent systems, and connected environments.

Core LearnLabs include:

- Robotics Lab (ROB) – working with robotics and intelligent systems
- Extended Reality Lab (RITA) – simulating and analyzing industrial settings in virtual “digital twins”
- Digital Production Lab (HANS) – exploring production systems and digital industrial processes

Key technologies and methods include:

- Robotics, especially ROS
- Machine learning (reinforcement learning, supervised learning, foundation models)
- IoT, including sensors, actuators, edge devices, and communication (MQTT)
- Predictive maintenance
- Supply chain management
- Digital twins
- Business processes

Together, these labs and technologies help you understand how computing can be applied to make industrial systems smarter, more connected, and more sustainable. Future Industries prepares you for careers where the goal is to apply computing to industrial production, robotic environments, innovation, and connected systems. What makes these paths attractive is that they are closely linked to how industry is changing today. Manufacturing, logistics, and industrial systems are becoming more digital, data-driven, and interconnected. In other words, you are not only learning how to work with computing technologies, you are learning how to apply them in industrial contexts where efficiency, sustainability, and innovation matter.



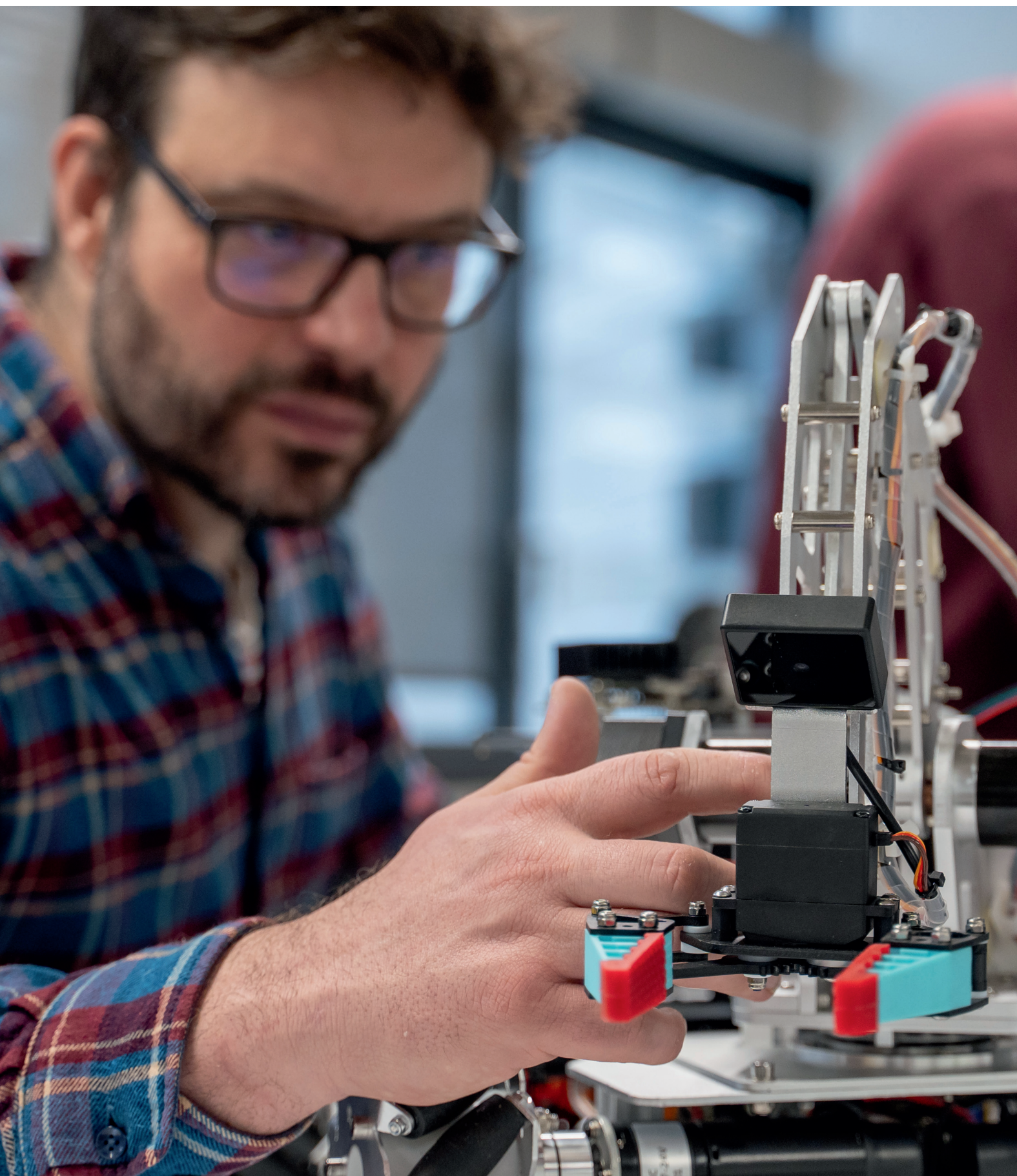
Career Prospects

This specialization is especially relevant for roles such as Robotics Engineer, IoT Specialist, Project Manager, Requirements Engineer, Software Engineer/Architect, Digital Manufacturing Specialist, Predictive Maintenance Manager, Supply Chain Manager, Digital Twin Specialist, or Industrial Innovation Consultant. It is a strong fit for students who want to continue into either industry or academia, as the program highlights both practical and methodological competencies in a project-based learning environment.

“We focus on developing innovative interaction concepts for AI and computing, ensuring that future digital systems evolve from being mere tools to becoming valuable team players.”



Philipp Wintersberger
Professor of Intelligent User Interfaces



APPLICATION AND CONTACT

You want to study at IT:U?

Are you ready to apply and start transforming tomorrow? We can't wait to receive your application!

Please fill in the IT:U application form and upload your files:

- Curriculum vitae (in English only)
- Proof of ID (passport/national ID card and Austrian or other EU residency card, if applicable)
- Academic certificates (Bachelor's; in original language and English or German translation/legalized, if applicable)
- Academic transcripts (Bachelor's; in original language and English or German translation/legalized, if applicable)
- Motivational letter (in English only, max. 1 page/approx. 500 words)

Tell us...

- A bit about who you are – your background and your future goals
- Why you are motivated to study at IT:U, and which of the four specializations (Designing Interactions; Digital Earth, Society, and Networks; Digital Humanities, Health, and Life Sciences; Future Industries) you are interested in
- Surprise us! If there's anything else you think we should know about your background, identity, interests, or talents, you can include it here

- Video submission/application video (in English only):

Requirements

- Must be 2-3 minutes in length. Submissions under 2 minutes or over 3 minutes will not be considered further
- Video submission – must include a video of yourself and speaking using your own voice

Tell us ...

- Why you are a good fit for the program and why the program is a good fit for you
- What you imagine to be the most interesting, challenging, and/or rewarding part of this program
- About an industrial, societal, and/or global challenge that you are passionate about, and how this program will help you address this challenge

Application period

The application is open year-round, but is split into two application cycles:

- January 1 through July 31 and August 1 through December 31.
- Applicants are welcome to apply once per application cycle.

Do you have any questions about the application?

Please feel free to reach out to our Student Administration Team.



APPLY
ONLINE



CONTACT:
Student Administration Team
study@it-u.at



YOU WANT TO FIND OUT MORE ABOUT STUDYING IN AUSTRIA?

If you are considering **STUDYING IN LINZ** then check out our short guide on the following pages! Find essential information about:

- Housing
- Campus Life
 - Food & Beverage
 - Sports
 - Other
- Discovering Linz

:Housing

Student Dormitories

Students have a variety of accommodation options available to them. Among the most popular choices are student dormitories, which are conveniently situated both near the campus and within the city.

Shared Apartment/Your Own Apartment

Another option is to rent a shared apartment or your own apartment. In Austria, it is quite common for students to live in a shared apartment (Wohngemeinschaft). Rent is more affordable, and you can easily meet new people and make friends.

Housing Options Upon Arrival

If you are unable to secure accommodation before arriving in Linz, you can consider temporarily renting an apartment through Airbnb or staying at a hostel/hotel.



:Campus Life

Food & Beverage

Science Café

With its snack bar concept, the Science Café aims to evoke a laid-back Mediterranean ambiance. Staff at the Science Café are skilled baristas, serving up Italian coffee specialties alongside a selection of Mediterranean-inspired breakfast and lunch options.

Die* Obelisk

Situated at the tram stop named JKU University, Obelisk offers a gateway to Austrian culinary delights for international students near the campus.

Teichwerk

Located by the pond, Teichwerk serves up imaginative, wholesome, and eco-conscious cuisine alongside refreshing beverages.

Cafeteria (Mensa JKU)

The Mensa cafeteria at JKU Linz offers a diverse selection of nutritious meals, including one vegetarian and one meat dish daily.

Ch@t-Café

Guests will indulge in fine coffee specialties, skillfully prepared by barista-trained staff. Additionally, an array of delightful cake variations and pastries sourced from local bakeries awaits.



Spar Digital Flagship Store

For those stomach rumbles before, between, or after lectures: Alongside deli meats, and bread, the campus branch also offers chilled beverages & smoothies, coffee to go, as well as fresh fruits and vegetables.

LUI – Linzer Uni Inn (Student Pub)

LUI is the cozy student hangout. In addition to a relaxed atmosphere, you will find a wide range of drinks and food at student-friendly prices.

KHG Cafeteria (Mensa)

In the KHG Mensa, you will find two daily menu choices: one featuring soup, a vegetarian main course, or a meat dish, each accompanied by optional salad.

Raab Cafeteria (Mensa)

Alongside its varied menu options, the restaurant offers coffee and drinks in the lounge.

Restaurants around Campus

If you are looking to grab a bite before your lectures, you will find several delightful restaurants conveniently located around the campus to choose from.

Sports**USI Courses**

The USI (University Sports Institute) offers a diverse range of sports and fitness activities.

Soccer Field and Volleyball Court

In addition to various fitness courses offered at the university, you can also book a soccer field or volleyball court to play with your friends.

Cycling

On campus, you can rent a bike via Nextbike App on your smartphone.

Library/Learning Center

The recently constructed JKU Learning Center boasts cutting-edge infrastructure, serving as a captivating hub for studying, collaboration, and research endeavors.

Other**Groceries & Cosmetic**

The most affordable supermarkets are Hofer and Penny, followed by Spar and then Billa. As for cosmetics, it can vary, but most people would say that DM is more affordable than Bipa. However Bipa offers a wider selection.

Pharmacies and Banks

Around campus you can find pharmacies as well as banks.

:Discovering Linz

Linz, the third-largest city in Austria, is a vibrant urban center that is rapidly evolving and expanding. Its dynamic growth makes it an exciting destination for international individuals, offering a wide range of opportunities, attractions, and cultural experiences. With its modern infrastructure, thriving economy, and rich cultural heritage, Linz is a welcoming city that attracts people from around the world. The city offers many opportunities for sports, culture, enjoying nature as well as gastronomy.

Top picks for Newcomers

- The Klangwolke
- The Pflasterspektakel
- The Krone Fest in Linz
- Bubble Days
- The Summer Open Air at the Donaupark Music Pavilion
- Christmas markets
- Urfahrner Markt
- Sofar Sounds
- The Linz Open
- The Ars Electronica Festival
- The Crossing Europe Film Festival
- The Stream Club Festival
- Posthof

**If you want to find out more,
check out our full Student Guide!**



STUDENT
GUIDE

Do you have any questions?

Please feel free to reach out. We are here to help you explore whether Linz could be your ideal home base!

CONTACT:

Student Support
student.support@it-u.at



IT:U Interdisciplinary Transformation University Austria

is the new technical university in Austria dedicated to digital transformation, driven by interdisciplinary research and project-based, personalized learning. Committed to inclusion and innovation, IT:U promotes lifelong learning and sustainable development, empowering students to make a positive impact globally.

Altenberger Straße 66c | Science Park 4 | 4040 Linz



IT-U.AT

IT:U interdisciplinary
transformation
university austria